

The Power of Vaccination

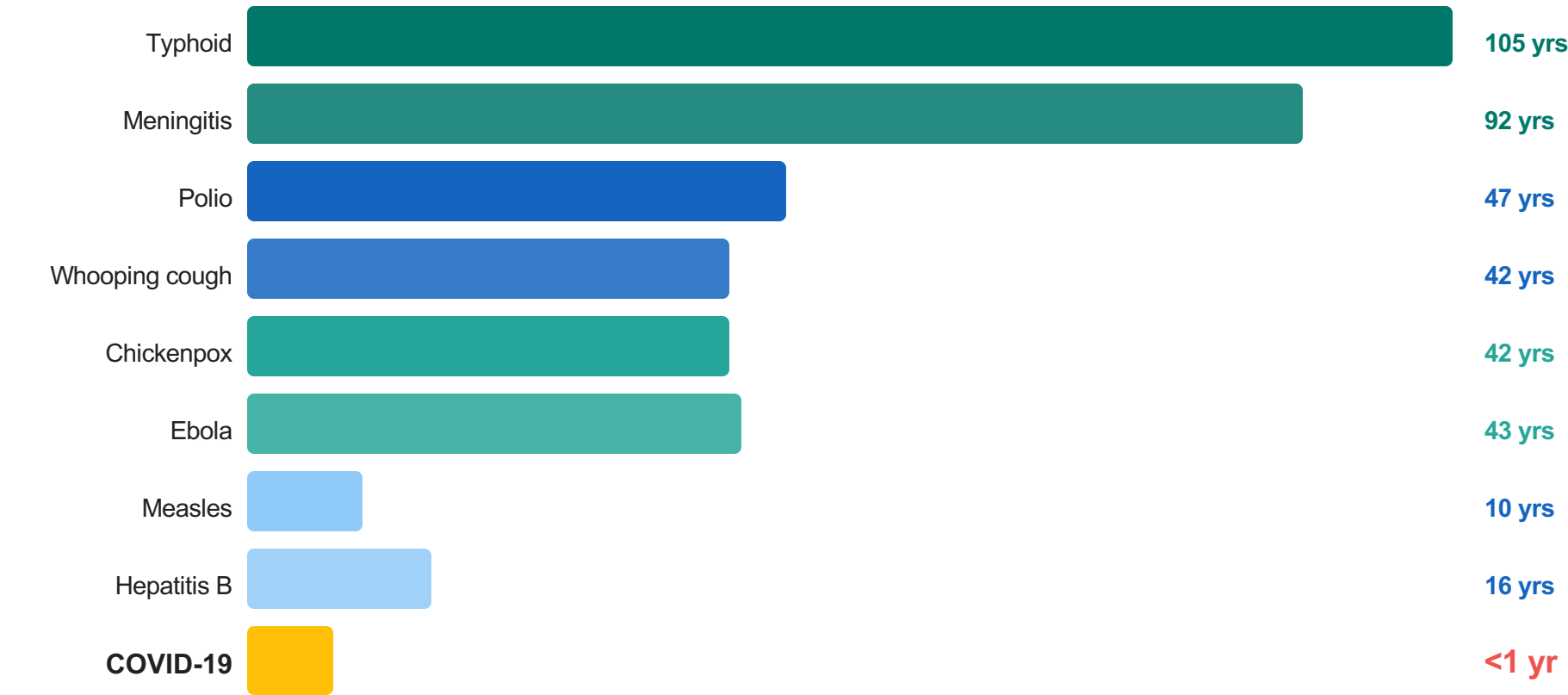
From Smallpox Eradication to COVID-19 in Record Time

1796–2020: Two Centuries of Vaccine Innovation

Source: Our World in Data

From 100 Years to Under 1 Year: Vaccine Development Is Accelerating

Time from pathogen identification to vaccine licensure (years)



Key insight: COVID-19's sub-1-year development is unprecedented in vaccine history

100% Case Reduction Achieved for Diphtheria, Polio, and Smallpox in the US

Disease	Pre-Vaccine Cases	Post-Vaccine Cases	Case Reduction	Pre-Vaccine Deaths	Death Reduction
Diphtheria	158	0	100%	13.7	100%
Measles	3,044	0.2	99.99%	2.5	100%
Smallpox	250	0	100%	2.9	100%
Acute Polio	141	0	100%	10	100%
Pertussis	1,534	52	96.6%	30.8	99.7%
Rubella	242	0.04	99.98%	0.09	100%
Hepatitis A	465	51	89%	0.5	88.7%
Acute Hepatitis B	273	44	83.9%	1	83.6%
Pneumococcal	233	139	40.5%	24	31.3%

100% Reduction (Eliminated)

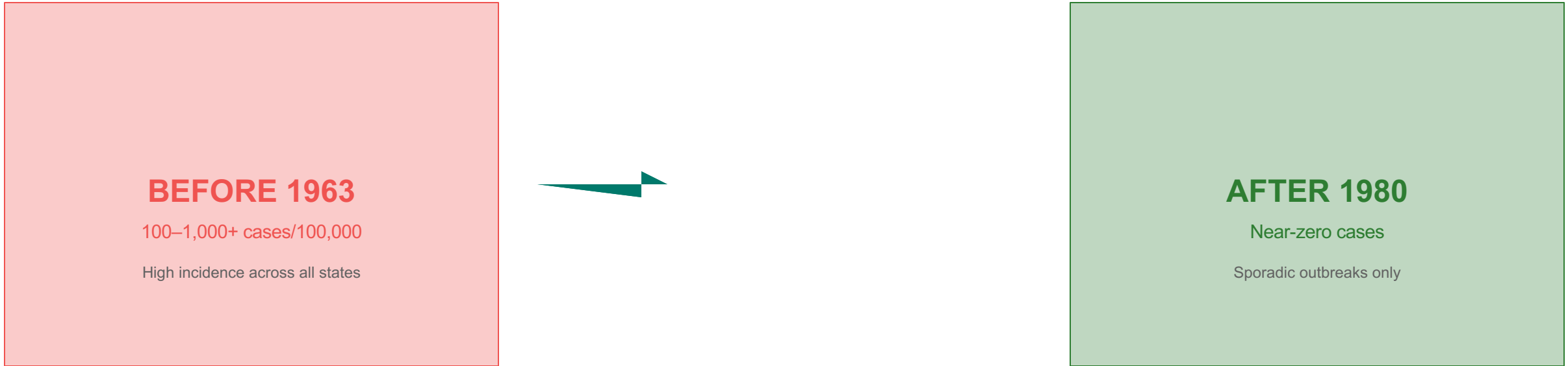
Near-Complete Reduction

Significant Reduction

Incomplete Reduction

Measles Cases Collapsed Across All 50 US States After Vaccine Mandates

Measles incidence per 100,000 population by state (1929–2022)



Key Milestones

1963

First measles vaccine
by John Enders

1971

MMR vaccine introduced
by Maurice Hilleman

1980

Kindergarten mandate
Cases drop to near-zero

Smallpox: The Only Human Disease Eradicated Through Vaccination



Why Smallpox Could Be Eradicated

1. Human-only reservoir

Virus infects only humans, no animal carriers

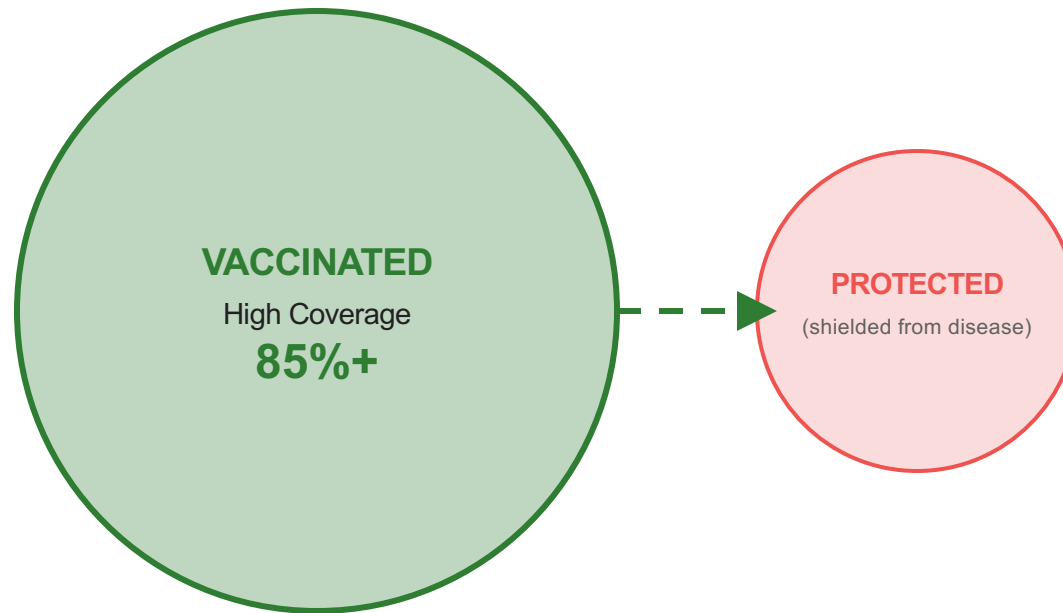
2. Distinguishable symptoms

Visible rash allows easy identification

3. Effective vaccine

Vaccinia virus vaccine was cheap and provided long-lasting immunity

Herd Immunity: How Vaccinating Many Protects the Most Vulnerable



Who Herd Immunity Protects

Newborns

Too young for vaccines

Immunocompromised

Cannot receive live vaccines

Cancer Patients

On chemotherapy

Elderly

Weakened immune response

Allergic Reactions

Contraindications to vaccines

Mechanism: When vaccination rate exceeds the threshold, pathogen transmission chains break, protecting those who cannot be vaccinated.

Gaps Persist: Pneumococcal Disease Shows Only 40% Case Reduction

Significant vaccine-preventable disease burden remains. Incomplete coverage and access gaps limit impact.

Pneumococcal Disease

40.5% case reduction

31.3% death reduction

Streptococcus pneumoniae causes pneumonia, meningitis, and bloodstream infections

Hepatitis A

89% case reduction

88.7% death reduction

Food/water transmitted liver infection, outbreaks in areas with limited sanitation

Acute Hepatitis B

83.9% case reduction

83.6% death reduction

Blood-borne liver disease, still affecting hundreds of thousands globally

Barriers to Complete Protection

Access Gaps

Geographic/economic barriers to vaccination

Supply Shortages

Manufacturing and distribution challenges

Trust Challenges

Vaccine hesitancy and misinformation

Strain Evolution

New serotypes bypassing current vaccines

Key Takeaways: Vaccines Have Eliminated Deadly Diseases — But the Job Isn't Done

Key Findings

1

100% Elimination Achieved

Diphtheria, smallpox, and both forms of polio eliminated in the US

2

Development Timelines Collapsed

From 100+ years (typhoid) to under 1 year (COVID-19)

3

Measles Near-Zero Nationally

Cases dropped to near zero across all 50 states after 1980 mandate

4

Smallpox: Only Eradicated Disease

Certified eradicated in 1980 after global vaccination campaign

5

Gaps Persist

Pneumococcal (40.5%), Hepatitis A (89%), Hepatitis B (83.9%) — need improved access, coverage, and trust

Call to Action for Global Health Systems

1. Strengthen vaccine supply chains and distribution networks
2. Improve access in underserved and remote communities
3. Build public trust through transparent communication and community engagement
4. Invest in next-generation vaccines for remaining diseases